Outcomes-based course syllabus

COURSE INFORMATION					
Course Code	**CMPE 103**	Course Title	**OBJECT-ORIENTED PROGRAMMING**	Course Credit	2 Units
Course Description	Introduces the fundamental concepts of programming from an object-oriented perspective. Topics are drawn from classes and objects, abstraction, encapsulation, data types, calling methods and passing parameters, decisions, loops, arrays and collections, documentation, testing and debugging, exceptions, design issues, inheritance, and polymorphic variables and methods. The course emphasizes modern software engineering and design principles.				
Pre-Requisites	CMPE 102 (PROGRAMMING LOGIC AND DESIGN)			Co-Requisites	N/A

VISION	PUP: The National Polytechnic University (PUP: Pambansang Politeknikong Unibersidad)
MISSION	Ensuring inclusive and equitable quality education and promoting lifelong learning opportunities through a re-engineered polytechnic university by committing to: ✓ provide democratized access to educational opportunities for the holistic development of individuals with global perspective ✓ offer industry-oriented curricula that produce highly skilled professionals with managerial and technical capabilities and a strong sense of public service for nation building ✓ embed a culture of research and innovation ✓ continuously develop faculty and employees with the highest level of professionalism ✓ engage public and private institutions and other stakeholders for the attainment of social development goal establish a strong presence and impact in the international academic community
QUALITY POLICY STATEMENT	The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of university services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.
	1. Creative and Critical Thinking. Graduates use their imaginative as well as a rational thinking ability to life situations in order push boundaries, realize possibilities, and deepen their interdisciplinary and general understanding of the world. 2. Effective Communication. Graduates are proficient in the four macro skills in communication (reading, writing, listening, and speaking) and are able to use these skills in solving problems. Making decisions, and articulating thoughts when engaging with people in various circumstances.

INSTITUTIONAL LEARNING OUTCOMES (ILO)	3. Strong Service Orientation. Graduates exemplify the potential of an efficient, well-rounded and responsible professional deeply committed to service excellence. 4. Community Engagement. Graduates take an active role in the promotion and fulfillment of various advocacies (educational, social and environmental) for the advancement of community welfare. 5. Adeptness in the Responsible Use of Technology. Graduates demonstrate optimized use of digital learning abilities, including technical and numerical skills. 6. Passion to Lifelong Learning. Graduates are enabled to perform and function in the society by taking responsibility in their quest to know more about the world through lifelong learning. 7. High Level of Leadership and Organizational Skills. Graduates are developed to become the best professionals in their respective disciplines by manifesting the appropriate skills and leaderships qualities. 8. Sense of Personal and Professional Ethics. Graduates show desirable attitudes and behavior either in their personal and professional circumstances. 9. Sense of National and Global Responsiveness. Graduates' deep sense of national compliments the need to live in a global village where one's culture and other people culture are respected.
COLLEGE GOALS	1. Provide quality education through instruction, advance research, and extension services. 2. Produce world-class professionals as potential industry leaders and job providers. 3. Develop and produce facilities through the use of adapted technology and indigenous materials. 4. Maintain, upgrade or improve facilities through the applications of engineering technology.

PROGRAM LEARNING OUTCOMES (PLO)

	Alignment to ILOs								
	1	2	3	4	5	6	7	8	9
1. Use an integrated development environment to write, compile, run and test simple object-oriented Python programs	✓				✓				
2. Identify classes, objects, members of a class and relationships among them	✓				✓				
3. Demonstrate the concepts of Abstraction and Encapsulation.	✓				✓				
4. Demonstrate the concepts of polymorphism and inheritance	✓				✓				
5. Solve real world problems using OOP techniques.	✓				✓				
6. Implement programs using Python features such as composition of objects, overloading, inheritance, Polymorphism etc.	✓				✓				
7. Develop and understand exception handling, multithreaded applications and GUI-based application Development	✓				✓				
8. Develop application project beneficial to society.				✓				✓	✓
9. Lead a group of students effectively to accomplish application programming activities and projects.							✓	✓	✓

PERFORMANCE INDICATORS (PI)		Alignment to PLOs								
		1	2	3	4	5	6	7	8	9
	• Explain and relate basic concepts of object-oriented programming (OOP) including encapsulation, classes, instances, attributes, methods, and constructors. • Explain and relate advanced concepts of OOP including inheritance, polymorphism, multiple inheritance, static methods, and object introspection. • Implement OOP principles in a number of training classes and a simplified version of Python Turtle Graphics.	✓								
	• Define terms related to object-oriented programming including object-oriented programming, class, instantiation, class library, instance or member variables, instance or member methods, public, protected, private, super class, parent class, base class, subclass. • Show how to create a class interface file. • Show how to create a class implementation		✓							
	• Appreciate the importance of maintaining abstraction barrier in software development • Understand the differences between statically and dynamically typed languages • Understand the concepts of object-oriented programming, including encapsulation, data hiding, fields and methods, • constructors, mutators/accessors, classes and objects, and their purposes of introducing them as a method of programming			✓						
	• Discuss Inheritance • Identify the types or level of inheritance • Describe the concept of inheritance in Object-oriented Programming • Describe Polymorphism • Discuss the different techniques in implementing Polymorphism in Object-oriented programming				✓				✓	
	• Discuss the principle of Exception handling in Python • Discover the different types of exceptions • Implement Exception handling in Python using the try and catch approach					✓				

	• Differentiate method overload and overriding • Discuss the concept of operator overloading and overriding • Write code that illustrate the usefulness of method overloading and overriding						✓			
	• Discuss how to use package and modules in Python • Explore the GUI-based Programming in Python • Discuss the concept of Event-driven programming							✓		
	• Apply knowledge and understanding to develop applications that could make impact to everyone lives. • Develop application beneficial to education sector								✓	
	• Perform leadership management to the group, assign tasks to the members, provide updates, milestones, progress of the class activity or group project.								✓	✓
COURSE LEARNING OUTCOMES **(CLO)**		**Alignment to PLOs**								
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
	1. Recognize the theoretical concept of Object-Oriented approach (class, objects, encapsulation, abstraction, polymorphism, Inheritance etc.)	✓			✓					
	2. Explain practical implementation of Object-Oriented Programming using Python and compare between Object Oriented Programming approach and procedural programming approach.	✓				✓				
	3. Apply the concept of Object-Oriented Programming using Python like class, objects, constructors & Destructors, Method Overloading, Operator Overloading, inheritance, Exception Handling, File Handling		✓							
	4. Create and execute different programs using Object Oriented Programming method.		✓	✓	✓	✓	✓	✓	✓	
	5. Develop simple GUI interfaces for a computer program to interact with users, and to understand the event-driven GUI handling principles using Python GUI Libraries.	✓	✓			✓				
	6. Demonstrate the use of various OOPs concepts with the help of programs.		✓		✓	✓	✓			

	7. Use dynamic and static polymorphism to process objects depending on their class.		✓	✓	✓	✓	✓	✓	✓	
	8. Understand and implement concepts on file streams and operations in Python programming for a given application programs.				✓					
	9. Produce robust and secure programs using exception handling and extensive program testing.			✓	✓	✓	✓	✓	✓	

OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Learning Content/Topics	Desired Learning Outcomes (DLOs)	Instructional Delivery Design			Assessment	Alignment to CLOs								
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)											
				Synchronous	Asynchronous		1	2	3	4	5	6	7	8	9
	a. Introduction to the course b. Introduction (Faculty and Students) c. Discussion of Course Syllabus including Grading System and General rules. d. Classroom Management	a. Demonstrate an understanding of what the subject is all about, what will be in scope for the semester, and what students are expected to learn. b. Communicate with fellow students and teacher and begin to establish rapport. c. Identify and explain the course assessment and validation criteria, including grading system to understand how to pass the subject. d. Explain what are the do's and don'ts while the class is on-going	Discussion of course learning content/topics, grading system, required software tools, assessments, activities, project groupings, grading system, class rules, and introductions	Discussion of course learning content/topics, grading system, required software tools, assessments, activities, project groupings, grading system, class rules, and introductions via zoom meeting or MS Teams											
1	Working with Python's String	**Module 1:** a. Describe the function of string operators. b. Use string handling techniques. c. Use for loops with string operations. d. Access individual characters in a string and apply basic string operations. e. Search and retrieve a substring from a string. f. Use string methods to manipulate strings.	Lecture Demonstrate example codes. Hands-on exercise/activity	Lecture Demonstrate example codes. Hands-on exercise/activity	Watch online videos of lecture/demonstration. Work with hands-on activities and submit with error-free source codes on your www.onlinegdb.com account.	Evaluate the output based on the rubrics including the design and codes.	✓	✓	✓						

2-3	File Handling in Python	**Module 2:** *a.Understand the structure of file systems* *b.Discuss opening files with different modes* *c.Introduce files as another kind of sequence that one can iterate over* *d.Explore the read/transform/write pattern*	Lecture Demonstrate example codes. Code some applications relating to the use of File handling Hands-on exercise/activity	Lecture Demonstrate example codes. Hands-on exercise/activity	Watch online videos of lecture/demonstration. Work with hands-on activities and submit with error-free source codes on your www.onlinegdb.com account.	Evaluate the output based on the rubrics including the design and codes.			✓	✓	✓	✓	✓	✓	✓
4- 5	Exception Handling in Python	**Module 3:** a. Describe what errors and exceptions are b. Handle errors and exceptions when they occur c. Define and use your own custom exceptions	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of Exception handling Hands-on exercise/activity	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of OOP. Hands-on exercise/activity	Watch online videos of lecture/demonstration. Work with hands-on activities and submit with error-free source codes on your www.onlinegdb.com account.	Evaluate the output based on the rubrics including the design and codes.			✓	✓	✓	✓	✓	✓	✓
6	Introduction to Object-oriented Programming	**Module 4:** a.Describe the elements and principles of object-oriented programming b.Apply the object-oriented concepts to software design c.Design and develop object-oriented computer programs d.Formulate problems as steps so as to be solved systematically	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of classes and objects. Hands-on exercise/activity	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of Linked List Hands-on exercise/activity	Watch online videos of lecture/demonstration. Work with hands-on activities and submit with error-free source codes on your www.onlinegdb.com account.	Evaluate the output based on the rubrics including the design and codes.			✓	✓	✓	✓	✓	✓	✓
7 -8	Abstraction and Encapsulation	**Module 5:** a.Describe the concept of encapsulation as a feature of the object-oriented programming style. b.Describe the concept of abstraction as a feature of the object-oriented programming style. c.Differentiate between encapsulation and abstraction by demonstrating their implementation	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of Stacks.	Watch online videos of lecture/demonstration. Work with hands-on activities and submit with error-free source codes on your www.onlinegdb.com account.	Evaluate the output based on the rubrics including the design and codes.			✓	✓	✓	✓	✓	✓	✓

			Abstraction and encapsulaton Hands-on exercise/activity	Hands-on exercise/activity										
9	MIDTERM WEEK													
10-11	Inheritance and Polymorphism	**Module 6:** a. Discuss the concept of inheritance b. Identify what are multiple inheritances c. Explore inheritance hierarchy d. Understand the concept of polymorphism e. Discuss the importance of polymorphism and inheritance f. Develop codes that implements inheritance and polymorphism in Python	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of Inheritance and Polymorphism Hands-on exercise/activity	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of Queues and Dequeues. Hands-on exercise/activity	Watch online videos of lecture/demonstration. Work with hands-on activities and submit with error-free source codes on your www.onlinegdb.com account.	Evaluate the output based on the rubrics including the design and codes.			✓	✓	✓	✓	✓	✓	✓
12-13	Overloading and overriding	**Module 7:** a. Discuss the importance of overloading and overriding b. Differentiate Method Overloading from Method Overriding c. Differentiate Operator Overload and Operator Override d. Develop codes that implements overloading and overriding	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use Overloading and overriding Hands-on exercise/activity	Demonstrate example codes using PyCharm IDE Code some applications relating to the use of Binary trees Hands-on exercise/activity	Watch online videos of lecture/demonstration. Work with hands-on activities and submit with error-free source codes on your www.onlinegdb.com account.	Evaluate the output based on the rubrics including the design and codes.			✓	✓	✓	✓	✓	✓	✓

14	Modules and Packages	**Module 8:** a. Describe what Python modules are and create your own b. Describe what Python packages are and create your own c. Demonstrate a code with the built-in Python modules **d.** Describe the file object in Python	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of modules and packages in OOP Hands-on exercise/activity	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of sorting and searching algorithm Hands-on exercise/activity	Watch online videos of lecture/demonstration. Work with hands-on activities and submit with error-free source codes on your www.onlinegdb.com account.	Evaluate the output based on the rubrics including the design and codes.			✓	✓	✓	✓	✓	✓	✓
15-16	Graphical User Interface and Event-driven Programming	**Module 9:** a. Differentiate between the functions of Windows applications and console applications b. Discuss the concept about graphical user interfaces c. Familiarize with some elements of good GUI design **d.** Explore event-handling procedures in Python by writing and registering event-handler methods	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of GUI-based event-driven programs Hands-on exercise/activity	Lecture Demonstrate example codes using PyCharm IDE Code some applications relating to the use of graphs. Hands-on exercise/activity	Watch online videos of lecture/demonstration. Work with hands-on activities and submit with error-free source codes on your www.onlinegdb.com account.	Evaluate the output based on the rubrics including the design and codes.			✓	✓	✓	✓	✓	✓	✓
17	APPLICATION PROJECT PRESENTATION	*Culminating activity given to the grouped students to test their mastery of the course by developing application programs utilizing all the theories and concepts acquired*	Project Presentation System Walk-through Simulation	Project Presentation System Walk-through Simulation	*Application Project Documentation* *Developed Application Program*	Evaluate the output based on the rubrics including the design and codes.	✓	✓	✓	✓	✓	✓	✓	✓	✓
18	**FINAL WEEK**														

LIST OF REFERENCES

1. Object-Oriented Python: Master OOP by Building Games and GUIs, Irv Kalb, No Starch Press, 2022
2. Python Object-Oriented Programming: Build Robust and Maintainable Object-Oriented Python Applications and Libraries, 4th Edition, Steven F. Lott, 2021
3. Python 3 Object Oriented Programming, Dusty Phillips, PAKT Publishing, 2021
4. Mastering Object-Oriented Python: Build powerful applications with reusable code using OOP design patterns and Python 3.7, 2nd Edition, 2019

CLASSROOM POLICY (to be filled out by the assigned faculty)

FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
1. The course is expected to have minimum 8 hands-on activities.	1. Attendance and Tardiness a. Classes start promptly on designated/agreed time

2. Activities are accomplished individual or in group mode, depends on the required output. 3. Evaluation/checking of output will be based on mobile application rubrics. 4. Assignments will be given throughout of the semester. Such requirements shall be due as announced in the class. 5. Students are required to attend classes regularly, including possible make-up classes. The student will be held liable for all topics covered and assignments made during his/her absence. The university guidelines on attendance and tardiness will be implemented. 6. Any evidence of copying or cheating during any examinations may result in a failing grade from the examination for all parties involved. Note that other universities guidelines shall be used in dealing with this matter. 7. Students are advised to keep graded work until the semester has ended. 8. No foods, drinks, cigarettes nor children are allowed inside the lecture and laboratory rooms. 9. Withdrawal and dropping from the subject should be done in accordance with existing university policies and guidelines regarding the matter	b. Attendance will be checked regularly 2. Internet Connection a. Internet connection may be interrupted from time to time. Students are encouraged to immediately return back to the meeting as soon as the connection has been restored. 3. All materials including video recording will be uploaded to our LMS 4. During Lectures a. Short discussion will follow at end of each lecture to check student understanding of the topic. b. Students will be asked to read from the presentation from time to time c. Students are encouraged to take down notes from the presentation 5. GAD/PWD Considerations a. All, regardless of gender will be treated equitably, with respect b. Proper considerations will be accorded to those with physical limitations (Read also netiquette guidelines for online students)
COURSE REQUIREMENT/S	**GRADING SYSTEM**
	Class Standing 70% Exercises/Activities Summative Quizzes Recitation Written Report/Reflective Journal/Portfolio Midterm / Final Examinations 30% 100% (Midterm Grade + Final Term Grade)/2 = Final Grade
CLASS INFORMATION	**FACULTY INFORMATION**
Section:	Name of Faculty:
Time:	Consultation Time:
Room:	Office Tel. No./ Mobile Phone No.

Semester:	E-Mail Address:

Suggested Rubrics for Assessment (to be filled out by the assigned faculty)

Suggested rubrics for assessment:

A. Essay Rubrics
B. Portfolio Rubrics
C. Students shall be graded or marked in accordance with the following system based on the Student Handbook:

GRADE/MARK	PERCENTAGE/ EQUIVALENCE	DESCRIPTION
1.0	97-100	Excellent
1.25	94-96	Excellent
1.50	91-93	Very Good
1.75	88-90	Very Good
2.0	85-87	Good
2.25	82-84	Good
2.50	79-81	Satisfactory
2.75	76-78	Satisfactory
3.0	75	Passing
5.0	65-74	Failure
Inc.		Incomplete
W		Withdrawn

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
ENGR. JULIUS S. CANSINO, PCpE – January 5, 2023 **Faculty/Committee/Date**	**ENGR. JULIUS S. CANSINO, PCpE – January 5, 2023** **Chairperson/Date**	**DR. REMEDIOS G. ADO – January 5, 2023** **Dean/Date**	**DR. EMANUEL C. DE GUZMAN – January 5, 2023** **Vice President for Academic Affairs/Date**

Form No. _____
Revision No. _____
Date of Approval _____
Semester _____
Academic Year _____